

ELITE MATHEMATICS

Pearson Edexcel International A Level

# Pure 2

## Adaptive Classified

Chapter 07 · Sequences and Series

### Questions

Student book · questions with writing space

- 25 pages
- Past papers up to January 2026

PURE 2 | TOPIC 07

# Sequences & Series

Your classified route

**10**

QUESTIONS

**68**

MARKS

**2**

FAMILIES

**2**

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

## TOPIC 07 | FAMILY 01

# Unknown Count

**WHAT THIS FAMILY ASKS**

Recognise the unknown count demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE****V01** core family move**4 classified questions**

29 total marks

8. A sequence  $a_1, a_2, a_3, \dots$  is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$
$$a_1 = p - 3$$

where  $p$  is a constant.

- (a) Find an expression for  $a_2$  in terms of  $p$ , giving your answer in simplest form.

(1)

Adaptive unit 1 · WMA12-01\_2021\_Jan\_Q08

DR ESAM AHMED

8. A sequence  $a_1, a_2, a_3, \dots$  is defined by

$$a_{n+1} = 2(a_n + 3)^2 - 7$$

$$a_1 = p - 3$$

where  $p$  is a constant.

Given that  $\sum_{n=1}^3 a_n = p + 15$

(b) find the possible values of  $a_2$

(6)

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2. A sequence is defined by

$$u_1 = 6$$
$$u_{n+1} = ku_n + 3$$

where  $k$  is a positive constant.

(a) Find, in terms of  $k$ , an expression for  $u_3$

(2)

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2. A sequence is defined by

$$u_1 = 6$$

$$u_{n+1} = ku_n + 3$$

where  $k$  is a positive constant.

Given that  $\sum_{n=1} u_n = 117$

(b) find the value of  $k$ .

(3)

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11. A sequence  $u_1, u_2, u_3, \dots$  is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where  $a$  and  $b$  are constants.

(a) Find, in terms of  $a$  and  $b$ ,

(i)  $u_2$

(ii)  $u_3$

(2)

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11. A sequence  $u_1, u_2, u_3, \dots$  is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where  $a$  and  $b$  are constants.

Given

$$\bullet \sum_{n=1}^3 u_n = 153$$

$$\bullet b = a + 9$$

(b) show that

$$a^2 - 5a - 66 = 0$$

(3)

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11. A sequence  $u_1, u_2, u_3, \dots$  is defined by

$$u_{n+1} = b - au_n$$

$$u_1 = 3$$

where  $a$  and  $b$  are constants.

(c) Hence find the larger possible value of  $u_2$

(3)

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**In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.**

- (2)

(3)

7: In this question you must show all stages of your working.  
Solutions relying entirely on calculator technology are not acceptable.

(ii) An **arithmetic** series begins

$$850 + 843 + 836 + \dots$$

Given that the sum of the first  $n$  terms of this series is  $S_n$   
find the greatest possible value of  $S_n$

(4)

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## TOPIC 07 | FAMILY 02

# Sigma Sums

**WHAT THIS FAMILY ASKS**

Recognise the sigma sums demand and complete the required reasoning in a clear, exam-ready order.

**VARIANT ROUTE**

**V01** core family move

**6 classified questions**

39 total marks

8. (i) An arithmetic series has first term  $a$  and common difference  $d$ .

Prove that the sum to  $n$  terms of this series is

$$\frac{n}{2}\{2a + (n - 1)d\} \tag{3}$$

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(ii) A sequence  $u_1, u_2, u_3, \dots$  is given by

$$u_n = 5n + 3(-1)^n$$

Find the value of

(a)  $u_5$

**(1)**

(b)  $\sum_{n=1}^{59} u_n$

**(3)**

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3. A sequence  $a_1, a_2, a_3, \dots$  is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(a) (i)  $a_1$

(ii)  $a_2$

(iii)  $a_3$

(3)

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3. A sequence  $a_1, a_2, a_3, \dots$  is defined by

$$a_n = \cos^2\left(\frac{n\pi}{3}\right)$$

Find the exact values of

(b) Hence find the exact value of

$$\sum_{n=1}^{50} \left\{ n + \cos^2\left(\frac{n\pi}{3}\right) \right\}$$

You must make your method clear.

(4)

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2. A sequence  $u_1, u_2, u_3, \dots$  is defined by

$$u_1 = 3$$

$$u_{n+1} = 2 - \frac{4}{u_n}$$

(a) Find the value of  $u_2$ , the value of  $u_3$  and the value of  $u_4$

(3)

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2. A sequence  $u_1, u_2, u_3, \dots$  is defined by

$$u_1 = 3$$

$$u_{n+1} = 2 - \frac{4}{u_n}$$

(b) Find the value of

$$\sum_{r=1}^{100} u_r$$

(2)

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5. (i) Find the value of

$$\sum_{r=1}^{\infty} 6 \times (0.25)^r$$

(3)

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(ii) A sequence  $u_1, u_2, u_3, \dots$  is defined by

$$\begin{aligned} u_1 &= 3 \\ u_{n+1} &= \frac{u_n - 3}{u_n - 2} \quad n \in \mathbb{N} \end{aligned}$$

(a) Show that this sequence is periodic. (2)

(b) State the order of this sequence. (1)

(c) Hence find 
$$\sum_{n=1}^{70} u_n$$
 (2)

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1. The sequence  $u_1, u_2, u_3, \dots$  satisfies

$$u_{n+2} = 3u_{n+1} - 2u_n$$

Given that

- $u_1 = 7$

- $u_3 = 4$

(a) find the value of  $u_2$

(2)

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1. The sequence  $u_1, u_2, u_3, \dots$  satisfies

$$u_{n+2} = 3u_{n+1} - 2u_n$$

Given that

- $u_1 = 7$
- $u_3 = 4$

(b) find  $\sum_{r=1} (u_r + 2r)$

(3)

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2. A sequence of numbers  $u_1, u_2, u_3, \dots$  is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where  $k$  is a constant.

(a) Show that  $u_5 = 7$

4

(3)

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2. A sequence of numbers  $u_1, u_2, u_3, \dots$  is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where  $k$  is a constant.

Given that  $\sum_{r=1} u_r = 30$

(b) find the value of  $k$ .

150

(2)

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2. A sequence of numbers  $u_1, u_2, u_3, \dots$  is defined by

$$u_1 = 7$$

$$u_{n+1} = (-1)^n u_n + k$$

where  $k$  is a constant.

(c) Hence find the value of  $\sum_{r=1} u_r$  (2)

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